MILITARY SPECIFICATION

NEBULIZER, MEDICINAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable manifold and nebulizer suitable for utilization with most Intermittent Positive Pressure Breathing (IPPB) apparatuses in conjunction with respiratory treatment and therapy.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping Fiberboard.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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DRAWINGS

Defense Personnel Support Center (DPSC)
DPSC Drawing No. 24348 Nebulizer, Medicinal, Plastic

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Materials. Materials shali be as specified in DPSC Drawing No. 24348. All materials shall be of the highest quality. Where not specified, materials used shall be suitable in all respects for the intended purpose.

3.3 Design and construction. The assembly shall consist of the manifold body, nebulizer, flexible corrugated main stream tube approximately 60 inches long with two supply lines, a corrugated flexible tube approximately 6 inches long, a mouthpiece, and an adapter. The components shall be fabricated in accordance with DPSC Drawing No. 24348. The manifold and nebulizer shall be universally adaptable for use with IPPB machines with either one or two supply lines. The manifold and nebulizer shall be furnished completely assembled, except for the mouthpiece, tubing adapter, short corrugated tubing and line adapter.

3.4 Performance. The nebulizer shall be suitable for use with IPPB apparatuses with either single or double supply lines. When tested as specified in 4.4.1, the nebulizer shall produce a very fine smoke-like mist. The observation of a heavy droplet-like mist or no mist at all during this test shall constitute failure. When tested as specified in 4.4.2, the system pressure should rise to 15 centimeters of water (cm H_2O) and then fall to between 0.5 to 3.0 cm H_2O. The test lung used during testing shall completely deflate during testing. There shall be no evidence of leakage during test.

3.5 Cleanliness. The entire nebulizer shall be provided in a clean, dry condition. The unit shall be free of dirt, hair, and any other foreign matter.

3.6 Instructions for use. Instructions for use shall be packaged with each nebulizer. Instructions shall include information for use with both one and two supply line machines as well as general information and cautionary notices.

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3.7 Workmanship. Workmanship shall be first class throughout. Units shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the nebulizer when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I. The unit for examination purposes shall be the individual nebulizer.

TABLE I. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Critical defects	II	1.0
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. All units shall be subject to the tests specified.

4.3 Examination. The nebulizers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in table II. Examination shall not be restricted to the classified possible defects listed below:

TABLE II. Classification of defects

Categories	Defects
Critical	
1	Nebulizer not free of dirt, hair, and any other foreign matter.
2	Siphon tube not free of kinks and bends.
3	Diaphragm not fabricated of silicon coated rubber.
Major	
101	Bag (packaging) not free of holes, rips, and tears.
102	Bag not sealed.
103	Tubing not free of holes and cuts.
104	Components not free of cracks and deformities.
Minor	
201	Tubing not secured in hose clips.
202	Instructions for use not included with unit.
203	Nebulizers not dry.
204	Manifold and nebulizer not assembled as required.

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4.3.2 Dimensional examination. The nebulizer shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the unit package, and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests.

4.4.1 Atomization test. A long length of vinyl tubing shall be connected to the piston located on top of the nebulizer assembly, just above the siphon tube connection. The other end of the tubing shall be connected to a source of clean, filtered, compressed air. The bottle shall be removed from the nebulizer. The end of the siphon tube shall be submerged a distance of one half inch for a period of 3 to 5 seconds in clean distilled water. The hole in the mist cap shall be observed for the condition of the mist produced. The mist shall be fine and smoke-like.

4.4.2 Operation test. The completely assembled nebulizer shall be attached to a single supply line IPPB machine (see 6.4). A rubber test lung shall be connected to the patient end of the manifold. The IPPB machine shall be set as follows:

a. Inspiratory pressure limit - 15 cm of water.
b. Expiratory time for apnea - 5 to 7 seconds.
c. Inspiratory sensitivity effort - 15 cm of water.
d. Air mix-off.

The respirator shall be connected to an oxygen tank and the flow rate set at 7 liters per minute. The system pressure gage shall be observed to ascertain proper operation. The pressure reading shall rise to 15 cm of water and then descent to 0.5 to 3.0 cm of water. If the unit fails to operate properly, it shall be checked for loose or defective components. Repaired units shall be retested.

5. PACKAGING

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Unit of issue. One package containing fifty individually packaged nebulizers, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Components comprising each nebulizer shall be thoroughly clean and packaged in a clear plastic bag having a wall thickness of not less than 2 mil. The bag shall be heat sealed or shall be designed with a friction lock type closure.

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5.1.2.2 Unit package. Fifty individual nebulizers shall be packaged in a box conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.1.3 Level B. Fifty nebulizers, individually packaged as specified in 5.1.2.1, shall be packed in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.4 Level C. Fifty individually packaged nebulizers shall be packaged in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2 Marking.

5.2.1 Individual package. Each package shall be marked in accordance with the contractor's commercial practice. Markings may be printed on the bag or on a label inserted within the bag. In addition, markings shall include the item name, name of contractor and the legend: "DISPOSABLE" or similar statement.

5.2.2 Unit package. Each unit package shall be marked as required for the exterior container specified in MIL-STD-129. The level of protection shall be shown on the exterior container a "A", "B" or "C" as applicable in lieu of "A/A", "B/B" or "C/C".

6. NOTES

6.1 Intended use. The nebulizer is intended for use with IPPB machines in conjunction with respiratory treatment and therapy.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging (see 5.1).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

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6.4 The Bird Corporation's Mark 7 Respirator, or equal, has been determined to be suitable for use in the test specified in 4.4.2.

6.5 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-528-4092	NEBULIZER, MEDICINAL, Plastic, 50's

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity
DSA-DM

Project No. 6515-1996

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-N-37455 - NEBULIZER, MEDICINAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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